

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 1: Pitot-Static Systems & Temperature Measurement

Syllabus Source: CAE Oxford (pp. 1-38) | Pass Target: >= 90% AviationExam

1. PRESSURE SOURCING: PITOT PRESSURE & STATIC PRESSURE

Aeronautical pressure instruments (Altimeter, ASI, VSI, Machmeter) rely on samples of atmospheric ambient pressure and impact pressure captured through dedicated external fuselage sensors:

- Static Pressure (P_s): The ambient barometric pressure of undisturbed still air at the aircraft's flight level. Sampled via flush static vents mounted on opposite sides of the fuselage (plumbing joined in a balance pipe to cancel yaw/sideslip errors).
- Pitot Pressure (P_t / Total Pressure): The total stagnation pressure captured by open-ended, forward-facing pitot probes aligned parallel to the longitudinal axis: Total Pressure (P_t) = Static Pressure (P_s) + Dynamic Pressure (q).
- Dynamic Pressure (q): Pressure resulting from aircraft forward velocity: $q = \frac{1}{2} \rho V^2$. Pitot pressure alone cannot measure airspeed; the static pressure must be subtracted (P_t - P_s = q)!
- Alternate Static Source: Located inside unpressurized flight deck or avionics bay. Due to airflow venturi suction around the fuselage, pressure inside the cabin is LOWER than ambient outside static pressure! Selecting alternate static causes: Altimeter reads HIGH, ASI reads HIGH, VSI shows momentary climb then stabilizes.

2. PITOT-STATIC ERRORS & SENSOR HEATING

Measurement errors distort raw pressure readings before reaching instrument capsules:

Error Source	Physical Mechanism & Flight Effect	Correction & Operational Mitigation
Instrument Error	Mechanical imperfections, capsule hysteresis, and friction in gear linkages.	Calibrated during factory manufacturing; negligible in modern digital ADC air data sensors.
Position / Pressure Error	Airflow around fuselage disturbs ambient static pressure at the vent.	Varies with Angle of Attack and airspeed/flaps. Sensed by ADC and corrected electronically via lookup tables.
Maneuver / Lag Error	Time delay as pressure transmits through long pneumatic tubing.	Eliminated in digital aircraft by mounting ADMs (Air Data Modules) right beside the pitot/static probes.
Electrical Heating	MANDATORY electrical resistive heating elements inside pitot probes and static plates.	Prevents ice blockage. Toggled on automatically on engine start or flight crew checklist. Inoperative probe heater risks catastrophic instrument misreadings!

[!] EXAM TRAP (AVIATIONEXAM) - Alternate Static Source Selection Flight Deck Effects

When the alternate static valve is opened on unpressurized aircraft: - Cabin static pressure is LOWER than true ambient atmospheric pressure (fuselage Bernoulli suction). - Altimeter: Reads HIGHER than actual altitude! - Airspeed Indicator (ASI): Reads HIGHER than actual airspeed! - Vertical Speed Indicator (VSI): Shows a MOMENTARY FALSE CLIMB, then returns to indicate true rate of climb/descent!

3. AIR TEMPERATURE MEASUREMENT: SAT, TAT & RAM RISE

Accurate outside temperature is essential for True Airspeed (TAS) computation, Mach number calculation, and icing detection:

- Static Air Temperature (SAT / OAT): The true ambient temperature of undisturbed outside air through which the aircraft is flying.
- Total Air Temperature (TAT): The temperature measured by a stagnation probe (Rosemount probe) exposed to the airstream. Due to adiabatic kinetic compression and boundary layer friction, air is brought to rest at the probe sensor, generating kinetic heating ('Ram Rise'): $TAT = SAT + \text{Ram Rise}$.
- Ram Rise Formula: $\text{Ram Rise} = (V_{TAS} / 100)^2$, or thermodynamically: $TAT = SAT \times (1 + 0.2 \times K_r \times M^2)$, where temperatures are in Kelvin ($K = ^\circ C + 273$), M = Mach number, and K_r = probe recovery factor (~1.0 for flight test).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): SAT and Ram Rise Calculation**

Exam Problem: Cruising at Mach 0.80, indicated TAT is -10 C (263 K). Recovery factor $K_r = 1.0$. What is true static air temperature (SAT)? Step 1: Formula: $TAT = SAT \times (1 + 0.2 \times M^2)$. Step 2: Mach factor: $1 + 0.2 \times (0.80)^2 = 1 + 0.128 = 1.128$. Step 3: Solve for SAT in Kelvin: $SAT_K = 263 / 1.128 = 233.15$ K. Step 4: Convert to Celsius: $SAT_C = 233.15 - 273.15 = -40$ C. Rule: TAT is ALWAYS WARMER than SAT in flight due to ram compression.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In an unpressurized aircraft, what occurs to the primary flight instruments if the pilot selects the alternate

- [A]: Altimeter reads low, ASI reads low, VSI shows momentary descent
- [B]: Altimeter reads HIGH, ASI reads HIGH, and VSI shows a momentary false CLIMB
- [C]: All instruments read zero
- [D]: Only the Machmeter is affected

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Airflow over the fuselage creates suction inside the cockpit, making cockpit pressure lower than outside static. Lower static pressure causes the altimeter to indicate higher, the ASI differential to indicate faster, and the VSI to sense a momentary climb.

Q2: An aircraft is cruising at Mach 0.80 with a Static Air Temperature (SAT) of -50°C. What is the approximate T

- [A]: -70°C
- [B]: -50°C

- [C]: -21°C
- [D]: 0°C

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: $TAT (K) = SAT (K) \times (1 + 0.2 \times M^2)$. Convert SAT: $-50^\circ C + 273 = 223 K$. Factor: $1 + 0.2 \times (0.80)^2 = 1 + 0.2 \times 0.64 = 1.128$. $TAT = 223 \times 1.128 = 251.5 K$. Convert back to Celsius: $251.5 - 273 = -21.5^\circ C (\sim -21^\circ C)$.

Q3: Which flight instruments require a pneumatic connection to the PITOT pressure line?

- [A]: Altimeter and VSI only
- [B]: Airspeed Indicator (ASI) and Machmeter only
- [C]: Altimeter, ASI and VSI
- [D]: Turn coordinator and artificial horizon

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Only the Airspeed Indicator and the Machmeter require pitot (total) pressure to measure dynamic pressure. The Altimeter and VSI connect exclusively to the static pressure line.